

Lab6: Insight into API and JSON

Objective

- Demonstrate the basic working knowledge of API and JSON
- Create back-end PHP APIs with MySQL and return JSON data
- Use JavaScript to consume APIs and handle JSON responses for front-end

Tips

- These labs are expected to take more than the allocated 2 hours. You should try the tasks before the lab. Practice makes perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

API Document (Standard Specification)

API (Application Programming Interface) is a set of rules that allows front-end and back-end systems to exchange data. In this lab, we use **JSON** as the standard data format. A complete API document includes:

- Interface name and description
- Request URL and HTTP method
- Request parameters
- Response format in JSON
- Error response

Example: Get Books by Author

- Description: Filter books by author name and return matching results.
- Basic information
 - Request URL: `getAPI.php`
 - Request Method: `GET`
 - Data Format: `application/json`
 - Authentication: Not required

- Request Parameters:

Name	Type	Required	Example	Description
author	string	No	Eric Matthes	Filter books by author name

- Success Response (JSON):

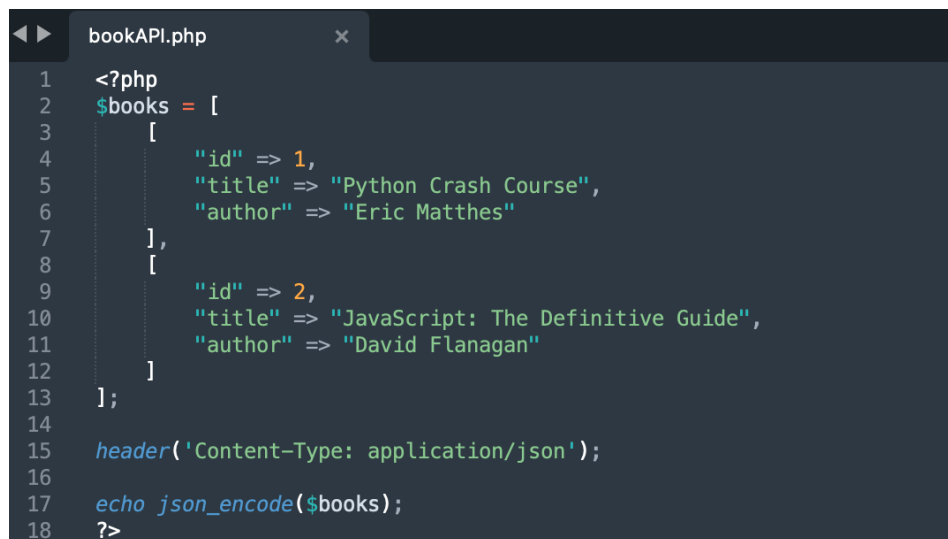
```
[
  {
    "id": 1,
    "title": "Python Crash Course",
    "author": "Eric Matthes"
  }
]
```

- Error Response

- No matching books: returns []
- Service unavailable: HTTP 404 or 500

A. Basics of API and JSON

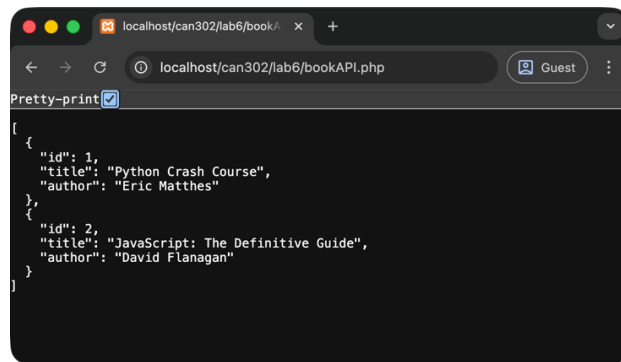
- Create a new subfolder "lab6" under "can 302"
- Create a new file "bookAPI.php" under "lab6" as follows:



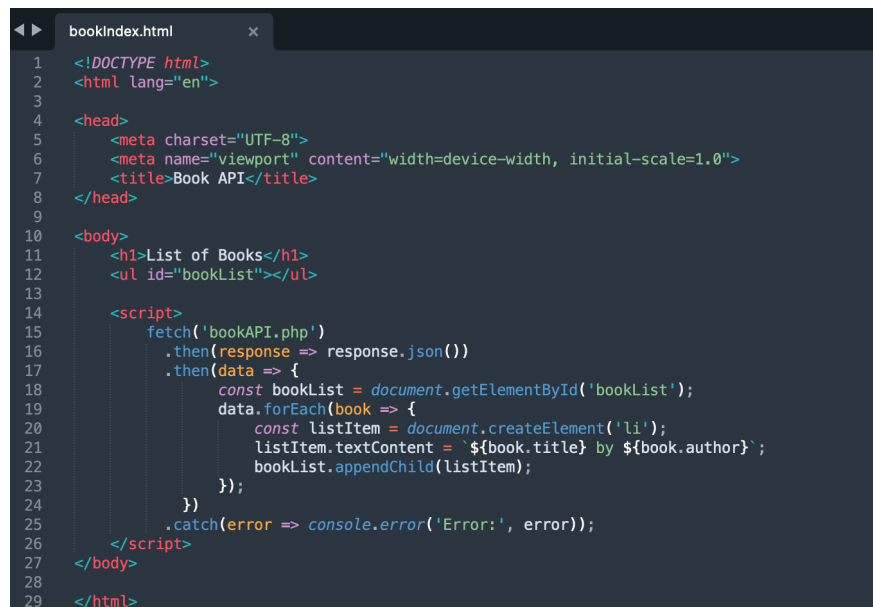
```
1 <?php
2 $books = [
3     [
4         "id" => 1,
5         "title" => "Python Crash Course",
6         "author" => "Eric Matthes"
7     ],
8     [
9         "id" => 2,
10        "title" => "JavaScript: The Definitive Guide",
11        "author" => "David Flanagan"
12    ]
13 ];
14
15 header('Content-Type: application/json');
16
17 echo json_encode($books);
18 ?>
```

In this PHP file:

- We first define an array of books as mock data
 - Then we set the response header to indicate that the content is in JSON format
 - Finally, we use `json_encode` to convert the array to a JSON string and echo it
- Access the webpage in your browser via URL: <http://localhost/can302/lab6/bookAPI.php>



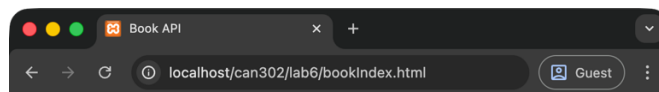
4. Create a new file "bookIndex.html" under "lab6" as follows:



In this HTML file:

- We have a simple HTML structure with a title and an empty unordered list
- We use the `fetch` API in JavaScript to make a GET request to our API
- When the response is received, we parse it as JSON and loop through the list of books. For each book, we create a new list item and append it to the unordered list

5. Access the webpage in your browser via URL: <http://localhost/can302/lab6/bookIndex.html>



List of Books

- Python Crash Course by Eric Matthes
- JavaScript: The Definitive Guide by David Flanagan

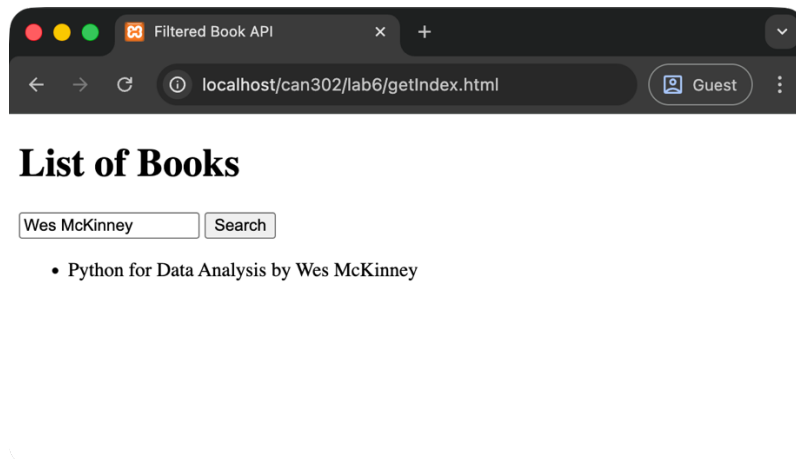
6. Create a new file "getAPI.php" under "lab6" as follows:

```
getAPI.php
1  <?php
2  $books = [
3      [
4          "id" => 1,
5          "title" => "Python Crash Course",
6          "author" => "Eric Matthes"
7      ],
8      [
9          "id" => 2,
10         "title" => "JavaScript: The Definitive Guide",
11         "author" => "David Flanagan"
12     ],
13     [
14         "id" => 3,
15         "title" => "Python for Data Analysis",
16         "author" => "Wes McKinney"
17     ]
18 ];
19
20 header('Content-Type: application/json');
21
22 if (isset($_GET['author'])) {
23     $author = $_GET['author'];
24     $filtered_books = array_filter($books, function ($book) use ($author) {
25         return strtolower($book['author']) === strtolower($author);
26     });
27     echo json_encode(array_values($filtered_books));
28 } else {
29     echo json_encode($books);
30 }
31 ?>
```

7. Create a new file "getIndex.html" under "lab6" to test the new API:

```
getIndex.html
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Filtered Book API</title>
8  </head>
9
10 <body>
11     <h1>List of Books</h1>
12     <input type="text" id="authorInput" placeholder="Enter author name">
13     <button onclick="fetchBooks()">Search</button>
14     <ul id="bookList"></ul>
15
16     <script>
17         function fetchBooks() {
18             const author = document.getElementById('authorInput').value;
19             let url = 'getAPI.php';
20             if (author) {
21                 url += '?author=' + author;
22             }
23             fetch(url)
24                 .then(response => response.json())
25                 .then(data => {
26                     const bookList = document.getElementById('bookList');
27                     bookList.innerHTML = '';
28                     data.forEach(book => {
29                         const listItem = document.createElement('li');
30                         listItem.textContent = `${book.title} by ${book.author}`;
31                         bookList.appendChild(listItem);
32                     });
33                 })
34                 .catch(error => console.error('Error:', error));
35         }
36     </script>
37 </body>
38
39 </html>
```

8. Access the webpage in your browser via URL: <http://localhost/can302/lab6/getIndex.html>



9. Create a new file "postAPI.php" under "lab6" as follows:

```

1 <?php
2 $books = [
3     [
4         "id" => 1,
5         "title" => "Python Crash Course",
6         "author" => "Eric Matthes"
7     ],
8     [
9         "id" => 2,
10        "title" => "JavaScript: The Definitive Guide",
11        "author" => "David Flanagan"
12    ]
13 ];
14
15 header('Content-Type: application/json');
16
17 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
18     $inputJSON = file_get_contents('php://input');
19     $input = json_decode($inputJSON, TRUE);
20     if (isset($input['author'])) {
21         $author = $input['author'];
22         $filtered_books = array_filter($books, function ($book) use ($author) {
23             return strtolower($book['author']) === strtolower($author);
24         });
25         echo json_encode(array_values($filtered_books));
26     } else {
27         echo json_encode(['error' => 'Author parameter is missing']);
28     }
29 } else {
30     echo json_encode(['error' => 'Only POST requests are allowed']);
31 }
32 ?>

```

10. Create a new file "postIndex.html" under "lab6" as follows:

```

1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7     <title>POST API</title>
8 </head>
9
10 <body>
11     <h1>List of Books</h1>
12     <input type="text" id="authorInputPost" placeholder="Enter author name">
13     <button onclick="fetchBooksPost()">Search</button>
14     <ul id="bookListPost"></ul>
15
16     <script>
17         function fetchBooksPost() {
18             const author = document.getElementById('authorInputPost').value;
19             const data = { author: author };

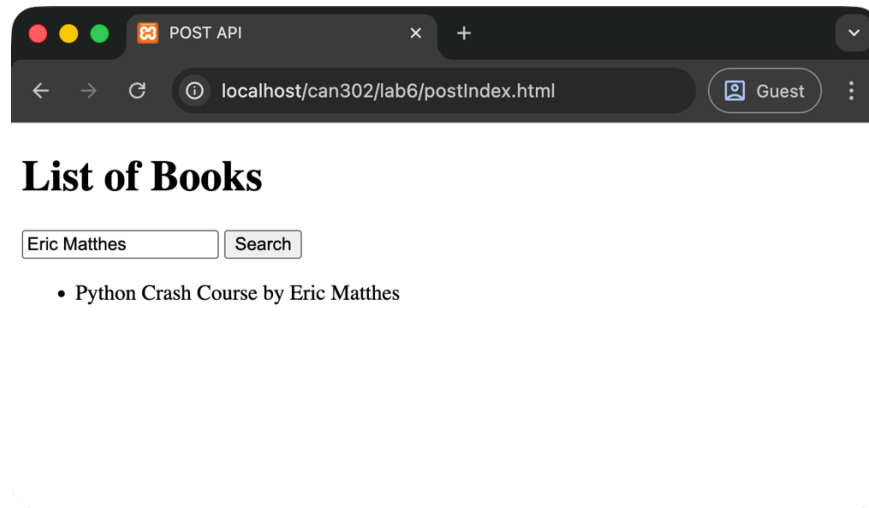
```

```

20     fetch('postAPI.php', {
21         method: 'POST',
22         headers: {
23             'Content-Type': 'application/json'
24         },
25         body: JSON.stringify(data)
26     })
27     .then(response => response.json())
28     .then(data => {
29         const bookList = document.getElementById('bookListPost');
30         bookList.innerHTML = '';
31         if (data.error) {
32             const errorItem = document.createElement('li');
33             errorItem.textContent = data.error;
34             bookList.appendChild(errorItem);
35         } else {
36             data.forEach(book => {
37                 const listItem = document.createElement('li');
38                 listItem.textContent = `${book.title} by ${book.author}`;
39                 bookList.appendChild(listItem);
40             });
41         }
42     })
43     .catch(error => console.error('Error:', error));
44 }
45 </script>
46 </body>
47 </html>

```

11. Access the webpage in your browser via URL: <http://localhost/can302/lab6/postIndex.html>



12. Import the following file to create a MySQL database named `library` with a table named `books` having columns `id`, `title`, and `author`:

```

dbSetup.sql
1  -- Create the database
2  CREATE DATABASE library;
3
4  -- Use the database
5  USE library;
6
7  -- Create the books table
8  CREATE TABLE books (
9      id INT AUTO_INCREMENT PRIMARY KEY,
10     title VARCHAR(100) NOT NULL,
11     author VARCHAR(100) NOT NULL
12 );

```

13. Create a new file "dbAPI.php" under "lab6" as follows:

```

dbAPI.php
1  <?php
2  $conn = new mysqli("localhost", "root", "", "library");
3  if ($conn->connect_error) {
4      die("Connection failed: " . $conn->connect_error);
5  }
6
7  header('Content-Type: application/json');
8
9  if ($_SERVER['REQUEST_METHOD'] === 'GET') {
10     $sql = "SELECT * FROM books";
11     $result = $conn->query($sql);
12
13     if ($result->num_rows > 0) {
14         $books = [];
15         while ($row = $result->fetch_assoc()) {
16             $books[] = $row;
17         }
18         echo json_encode($books);
19     } else {
20         echo json_encode([]);
21     }
22 } elseif ($_SERVER['REQUEST_METHOD'] === 'POST') {
23     $inputJSON = file_get_contents('php://input');
24     $input = json_decode($inputJSON, TRUE);
25     if (isset($input['title']) && isset($input['author'])) {
26         $title = $conn->real_escape_string($input['title']);
27         $author = $conn->real_escape_string($input['author']);
28         $sql = "INSERT INTO books (title, author) VALUES ('$title', '$author')";
29         if ($conn->query($sql) === TRUE) {
30             echo json_encode(['message' => 'Book added successfully']);
31         } else {
32             echo json_encode(['error' => 'Error adding book: ' . $conn->error]);
33         }
34     } else {
35         echo json_encode(['error' => 'Title and author are required']);
36     }
37 }
38
39 $conn->close();
40 ?>

```

This API interacts with a MySQL database:

- It can retrieve all the books from the `books` table using a `GET` request
- It can add a new book using a `POST` request

14. Create a new file "dbIndex.html" under "lab6" as follows:

```

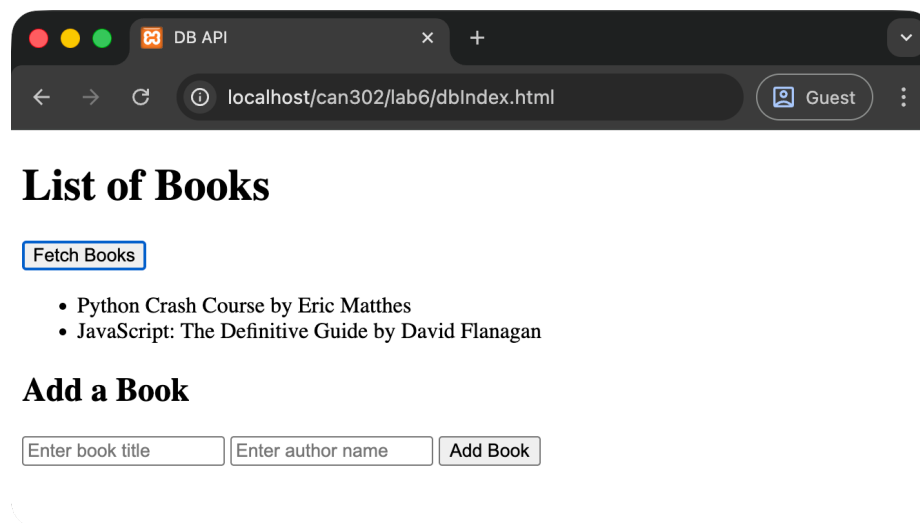
dbIndex.html
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>DB API</title>
8  </head>
9
10 <body>
11     <h1>List of Books</h1>
12     <button onclick="fetchBooksFromDB()">Fetch Books</button>
13     <ul id="bookListDB"></ul>
14     <h2>Add a Book</h2>
15     <input type="text" id="titleInput" placeholder="Enter book title">
16     <input type="text" id="authorInputDB" placeholder="Enter author name">
17     <button onclick="addBookToDB()">Add Book</button>
18

```

```
19 <script>
20   function fetchBooksFromDB() {
21     fetch('dbAPI.php')
22       .then(response => response.json())
23       .then(data => {
24         const bookList = document.getElementById('bookListDB');
25         bookList.innerHTML = '';
26         data.forEach(book => {
27           const listItem = document.createElement('li');
28           listItem.textContent = `${book.title} by ${book.author}`;
29           bookList.appendChild(listItem);
30         });
31       })
32     .catch(error => console.error('Error:', error));
33   }
34
35   function addBookToDB() {
36     const title = document.getElementById('titleInput').value;
37     const author = document.getElementById('authorInputDB').value;
38     const data = { title: title, author: author };
39     fetch('dbAPI.php', {
40       method: 'POST',
41       headers: {
42         'Content-Type': 'application/json'
43       },
44       body: JSON.stringify(data)
45     })
46     .then(response => response.json())
47     .then(data => {
48       if (data.error) {
49         alert(data.error);
50       } else {
51         alert(data.message);
52         fetchBooksFromDB();
53       }
54     })
55     .catch(error => console.error('Error:', error));
56   }
57 </script>
58 </body>
59
60 </html>
```

This HTML file has buttons to fetch books from the database and add a new book to the database. When a new book is added, it fetches the updated list of books

15. Access the webpage in your browser via URL: <http://localhost/can302/lab6/dbIndex.html>



B. Example: E-Commerce Product API

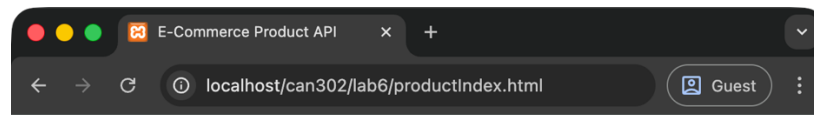
1. In this section, we will create a **product API** for an e-commerce system and use JavaScript to fetch and display product data. This example is fully aligned with Lab5.
2. Create a new file "productAPI.php" under "lab6". This file provides the API by reading product data from the "newStore" database and returning JSON format.

```
productAPI.php
1  <?php
2  // Create database connection
3  $conn = new mysqli("localhost", "root", "", "newStore");
4
5  // Set response header to JSON
6  header("Content-Type: application/json");
7
8  // Query product data
9  $sql = "SELECT * FROM products LIMIT 3";
10 $result = $conn->query($sql);
11
12 $products = array();
13 while($row = $result->fetch_assoc()) {
14     array_push($products, $row);
15 }
16
17 // Output JSON
18 echo json_encode($products);
19
20 $conn->close();
21 ?>
```

3. Create a new file "productIndex.html" to call the API using JavaScript `fetch()` and display product information.

```
productIndex.html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>E-Commerce Product API</title>
7  </head>
8
9  <body>
10     <h1>Product List (from API)</h1>
11     <ul id="productList"></ul>
12
13     <script>
14         // Fetch product data from API
15         fetch('productAPI.php')
16         .then(response => response.json())
17         .then(data => {
18             const productList = document.getElementById("productList");
19             data.forEach(product => {
20                 const li = document.createElement("li");
21                 li.textContent = product.name + " - $" + product.price;
22                 productList.appendChild(li);
23             });
24         })
25         .catch(error => console.error("Error:", error));
26     </script>
27 </body>
28
29 </html>
```

4. Access the webpage in your browser via URL: <http://localhost/can302/lab6/productIndex.html>



Product List (from API)

- MacBook - \$999.00
- Surface - \$999.00
- MateBook - \$999.00

C. Hand-on Exercises: Online Shop

Please update the online shopping website (Lab5) using APIs and JSON for data handling, which reflects real-world practices in the web development.

Summary

We have achieved the following things in this lab session:

- Write PHP-based APIs
- Use JSON for data exchange
- Update front-end to interact with APIs
- Access MySQL database through APIs